

Problem Set 10

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1 Questions and Responses

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Table 1: Optimal Tuning Parameters

Algorithm	Penalty	Estimate	Cost Complexity	Tree Depth	Min N	Hidden Units	Neighbors	Cost	RBF σ
Logit	0.00	0.85	—	—	—	—	—	—	—
Tree	—	0.87	0.00	15.00	10.00	—	—	—	—
Nnet	1.00	0.84	—	—	—	7.00	—	—	—
KNN	—	0.84	—	—	—	—	30.00	—	—
SVM	—	0.86	—	—	—	—	—	2.00	0.25

Out-of-sample accuracy ranges from 0.84 to 0.87 across the five algorithms. The decision tree performs best at 0.87, followed by the SVM at 0.86 and the LASSO logit at 0.85. The neural network and k-nearest neighbors tie for the weakest performance at 0.84.